PAPER_1453 — UQFF: Antimatter Production Max Efficiency (Bucket K)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket K **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — formal catalog tuple in Bucket K **Calculator surface:** `calculate_bsm_constraints({...})`

Closure

```
F_TRZ × β_i = 0.0603
```

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve this observable by different methods. Closure uses only canonical primitives. Zero free parameters.

Reference

- Catalog tuple resides in `_bsm_constraints_report()` or equivalent surface in `uqff_pure_calculator.py`.
- Related foundational papers: PAPER_646, PAPER_1156, PAPER_1203.
